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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CCT122 – REMEDIAL ENGINEERING

(For CCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

Unit – I

	* Marks	BT*	CO*	PO*
1. a) List and discuss any four causes of concrete deterioration.	10	L2	1	1
b) Explain Corrosion Mechanism in Reinforced Concrete.	10	L2	2	1
2. a) Explain carbonation test on concrete structures.	10	L2	1	1
b) Describe the effects of cover thickness and cracking on serviceability of concrete.	10	L2	2	2
3. a) What is nondestructive testing? Explain principle and procedure of Ultrasonic pulse velocity test.	10	L2	1	2
b) What are the methods adopted for reducing the initiation of corrosion in concrete structures, Explain any four methods?	10	L2	2	2

Unit – II

4. a) Briefly explain about:				
i. Sulphur infiltrated concrete				
ii. Ferro cement concrete	10	L2	3	1
b) Discuss any two methods of repairing cracks.	10	L2	4	1
5. a) Briefly explain about:				
i. Expansive cement				
ii. Fiber reinforced concrete	10	L2	3	1
b) Discuss any four applications of polymer concrete in repair of damaged buildings.	10	L2	4	1
6. a) Briefly explain about foamed concrete and vacuum concrete.	10	L2	3	1
b) Discuss Guniting and shotcrete with respect to features and applications.	10	L2	4	1

Unit – III

7. Write a short note on				
i. Case studies of any two retrofitting of buildings				
ii. Case studies of any two engineered demolition techniques for dilapidated structures	20	L2	5	1
8. Discuss RC Jacketing of Column members highlighting objectives, materials used, construction procedure, advantages and limitations.	20	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
